CDI COLLEGE / DATA SCIENCE TOOLS AND TECHNIQUES

Portfolio Technical Analysis Visualization Dashboard

BDA400 - Assignment 6 | Visualization Phase

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COURSE	BDA400
ASSIGNMENT	Assignment 6 - Technical Analysis using R, Visualization Phase

Project description

An interactive R Shiny dashboard for the five-symbol portfolio used in earlier project stages. It fetches daily historical OHLCV prices from Yahoo Finance with quantmod, groups them into daily, weekly or monthly bars, and displays line, candlestick or area charts with ggplot2. The dashboard offers independent SMA, RSI and MACD display controls and customizable indicator periods. Short and long moving-average crossings generate Buy/Sell annotations; all other bars are Hold.

Assignment repository

https://github.com/mahmuudbakkar-pixel/TechnicalAnalysis/tree/main/Assignment6

Files in the repository

app.R - complete, documented Shiny dashboard; **README.md** - setup and usage; **run_tests.R** and **test_results.txt** - offline verification. The dashboard includes a genuine CSV fallback for times when Yahoo Finance cannot be reached.

Open the public repository folder above to review and run the code. Live data availability depends on the Yahoo Finance connection.